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pressure of a mixture of non-reacting gases is the sum of partial pressures of the gases present in the mixture" where the partial pressure of a component gas is the pressure that it would exert if it were present alone in the same volume and temperature. This is known as *Dalton's law of partial pressures*.

i.e., for a mixture containing three gases 1, 2 and 3 with partial pressures p_1, p_2 and p_3 in a container with volume V, the total pressure P_{total} will be give by

$$P_{total} = p_1 + p_2 + p_3 \text{ ------- (6.12)}$$

Assuming that the gases behave ideally,

$$p_1 = n_1 \frac{RT}{V}; p_2 = n_2 \frac{RT}{V}; p_3 = n_3 \frac{RT}{V}$$

$$P_{Total} = n_1 \frac{RT}{V} + n_2 \frac{RT}{V} + n_3 \frac{RT}{V}$$

$$= (n_1 + n_2 + n_3) \frac{RT}{V}$$

$$P_{Total} = n_{Total} \left(\frac{RT}{V}\right) ------ 6.13$$

The partial pressure can also be expressed as

$\left(\frac{RT}{V}\right)$ can be expressed as $\frac{p_1}{n_1}$ or $\frac{p_2}{n_2}$ or $\frac{p_3}{n_3}$

or in general $\frac{p_i}{n_i}$

Therefore

$$P_{Total} = n_{Total} \frac{p_i}{n_i} = \frac{n_{Total}}{n_i} p_i$$

$$\Rightarrow p_i = \frac{n_i}{n_{Total}} P_{Total} = x_i P_{Total} \text{ ------- 6.14}$$

where x_i is the mole fraction of the ith component.

Application of Dalton's law

In a reaction involving the collection of gas by downward displacement of water, the pressure of dry vapor collected can be calculated using Dalton's law.

$$P_{\text{dry gas collected}} = p_{total} - p_{\text{water vapour}}$$

$p_{\text{water vapour}}$ is generally referred as aqueous tension and its values are available for air at various temperatures.

Let us understand Dalton's law by solving this problem. A mixture of gases contains 4.76 mole of Ne, 0.74 mole of Ar and 2.5 mole of Xe. Calculate the partial pressure of gases, if the total pressure is 2 *atm*. at a fixed temperature.

Solution:

$$p_{Ne} = x_{Ne} P_{Total}$$

$$x_{Ne} = \frac{n_{Ne}}{n_{Ne} + n_{Ar} + n_{Xe}}$$

$$= \frac{4.76}{4.76 + 0.74 + 2.5} = 0.595$$

$$x_{Ar} = \frac{n_{Ar}}{n_{Ne} + n_{Ar} + n_{Xe}}$$

$$= \frac{0.74}{4.76 + 0.74 + 2.5} = 0.093$$

$$x_{Xe} = \frac{n_{Xe}}{n_{Ne} + n_{Ar} + n_{Xe}}$$

$$= \frac{2.5}{4.76 + 0.74 + 2.5} = 0.312$$

$$P_{Ne} = x_{Ne} P_{Total} = 0.595 \times 2$$

$$= 1.19 \text{ atm.}$$